Demand Planning and Weather Sensitivity Guide

Document owner: Supply Planning
Department: Planning
Version: 2026.1
Effective date: 2026-01-01

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Purpose

This guide explains how TraceStock AI forecasts demand, which categories respond to weather and how strongly, and what evidence is required before a planner or an agent recommends a stock change. It exists because the largest forecasting errors in this business come from treating weather-sensitive categories as if they behaved like stable ones.

Forecasting Principle

Forecasts are built from recorded sales history first, adjusted for known seasonality second, and adjusted for forecast conditions third. Conditions never override history. A weather forecast changes the size of an expected movement; it does not create demand where the sales history shows none.

Every forecast must be reproducible. If a recommendation cannot be traced back to specific sales records over a stated period, it is an opinion and must be labelled as one.

Category Sensitivity Classes

Categories fall into three sensitivity classes.

Class A, strongly weather-driven. Demand can move by three times baseline or more within a week of a condition change.

- Air Conditioners: driven by sustained high temperature
- Space Heaters, driven by sustained low temperature
- Sunscreen, driven by high UV and clear conditions

Class B, moderately weather-driven. Demand moves noticeably but rarely more than doubles.

- Rain Jackets: driven by sustained precipitation
- Running Shoes, driven by mild, dry conditions and by seasonal transitions

Class C, weather-independent. Demand responds to promotions, launches, and general consumer cycles.

- Laptops, Smartphones, Headphones, Coffee Machines, Office Chairs

Class C categories should never be explained by weather. If Class C sales move sharply, look at pricing, promotions, stock availability, or a competitor event before considering anything else.

Weather Trigger Thresholds

The following thresholds are used as planning signals. They are indicative, not automatic ordering rules.

Category	Trigger condition	Sustained for	Expected effect
Air Conditioners	Daily maximum above 28°C	3 consecutive days	Demand 2x to 4x baseline
Air Conditioners	Daily maximum above 33°C	2 consecutive days	Demand 4x or more, stockout risk high
Space Heaters	Daily maximum below 5°C	3 consecutive days	Demand 2x to 3x baseline
Space Heaters	Daily maximum below 0°C	2 consecutive days	Demand 3x or more
Rain Jackets	Precipitation on 4 of 7 days	1 week	Demand 1.5x to 2x baseline
Sunscreen	UV index 6 or above with clear skies	3 consecutive days	Demand 2x to 3x baseline
Running Shoes	Daily maximum 12°C to 22°C, dry	5 consecutive days	Demand 1.3x to 1.6x baseline

Thresholds apply to the forecast for the region a site serves, not to a national average. Munich Fulfillment and Hamburg Warehouse regularly see different conditions in the same week, and planning them together produces poor results.

Regional Weather Differences

The four sites cover meaningfully different climates.

- **Hamburg Warehouse:** coastal, wettest of the four, mildest summers. Rain

Jackets over-index here year round. Air Conditioner demand is the lowest of the four sites.

- **Munich Fulfillment:** most continental, coldest winters and hottest

summers. Both Space Heaters and Air Conditioners over-index here, at opposite ends of the year.

- **Berlin Hub:** moderate on all dimensions. Used as the national baseline

when a single reference is needed.

- **Frankfurt DC:** mild winters, warm summers, lowest Space Heater demand.

When a weather-driven recommendation is made for one site, state the site. A recommendation to increase Air Conditioner stock is materially wrong if applied to Hamburg during the same heatwave that justifies it in Munich.

Seasonal Baselines

Independent of short-term weather, the following seasonal patterns hold.

- **January to February:** Space Heaters at annual peak. Electronics soft after the December period. Office Chairs rise with new-year home-office purchases.

- **March to April:** Running Shoes begin their climb. Rain Jackets peak in

many years. Space Heaters fall away quickly.

- **May to June:** Sunscreen and Air Conditioners begin rising. Running Shoes

at annual peak.

- **July to August:** Air Conditioners and Sunscreen at annual peak. Highest

stockout risk of the year across Class A categories.

- **September to October:** Transition period. Laptops rise with the academic

cycle. Rain Jackets rise again.

- **November to December:** Electronics at annual peak driven by promotional

events. Space Heaters begin rising. Highest overall revenue months.

Stockout Risk Assessment

Before recommending a stock increase, the following must be checked in order.

- Current stock on hand at the specific site
- The reorder point for that product at that site
- Recent sales velocity, measured over the trailing 14 and 30 days
- Supplier lead time for that product
- Forecast conditions over the lead-time window, for Class A and B categories

A stockout recommendation is only well founded when sales velocity multiplied by lead time exceeds current stock on hand. Recommending replenishment because stock looks low, without reference to velocity and lead time, is not acceptable.

Over-Ordering Risk

Weather-driven demand reverses as quickly as it appears. Class A categories carry real over-ordering risk, because a heatwave that ends leaves stock that will not sell for another year.

For this reason, weather-justified orders should not exceed the expected additional demand over the forecast window plus a two-week buffer. Ordering to cover an entire season on the strength of a five-day forecast is explicitly discouraged, and requires Supply Planning approval regardless of value.

Evidence Standards

Any demand recommendation must state:

- The product and the site it applies to
- The sales period examined and the velocity observed
- The current stock position and reorder point

- The lead time assumed
- For Class A and B categories, the forecast conditions relied on
- The expected duration of the effect

A recommendation missing the site or the period is not actionable and should be returned for completion rather than acted on.

Document Control

Field	Value
Version	2026.1
Effective date	2026-01-01
Last reviewed	2026-01-01
Next review due	2027-01-01
Owner	Supply Planning
Approved by	Head of Supply Planning
Classification	Internal

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